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발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

이번 주 진행사항

- Time-Frequency Pattern-Specific (TFPS)
    - 논문 리뷰

Time-Frequency Pattern-Specific

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Learning Pattern-Specific Experts for Time Series Forecasting Under Patch-level Distribution Shift

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Background

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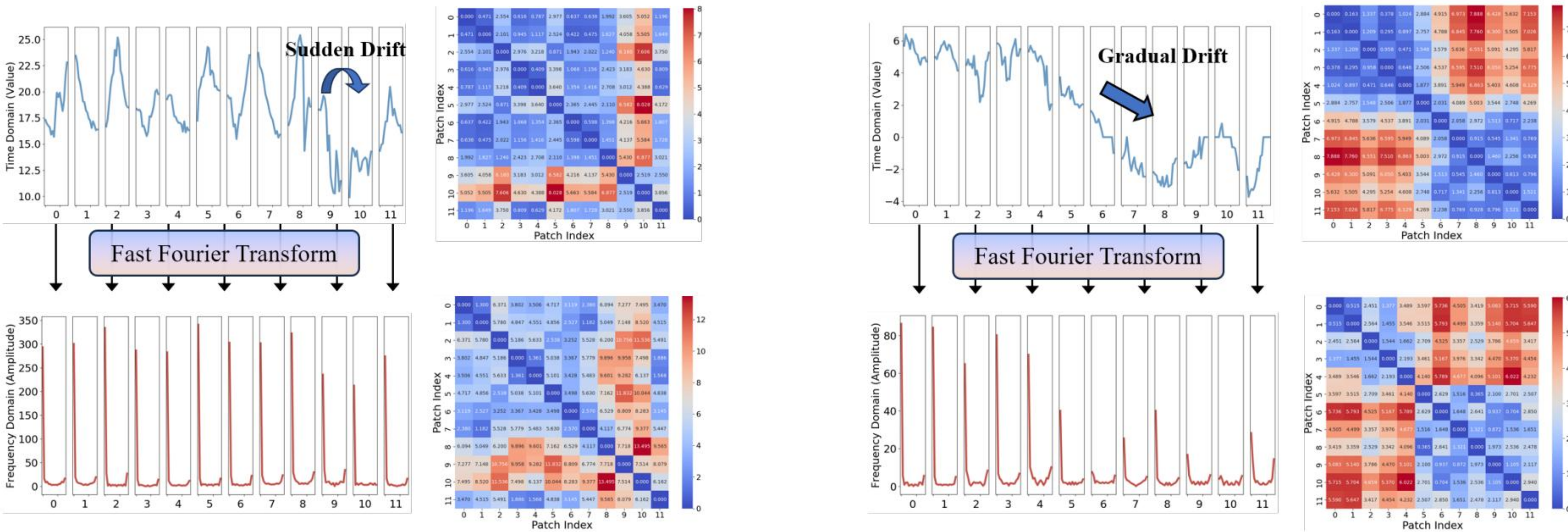
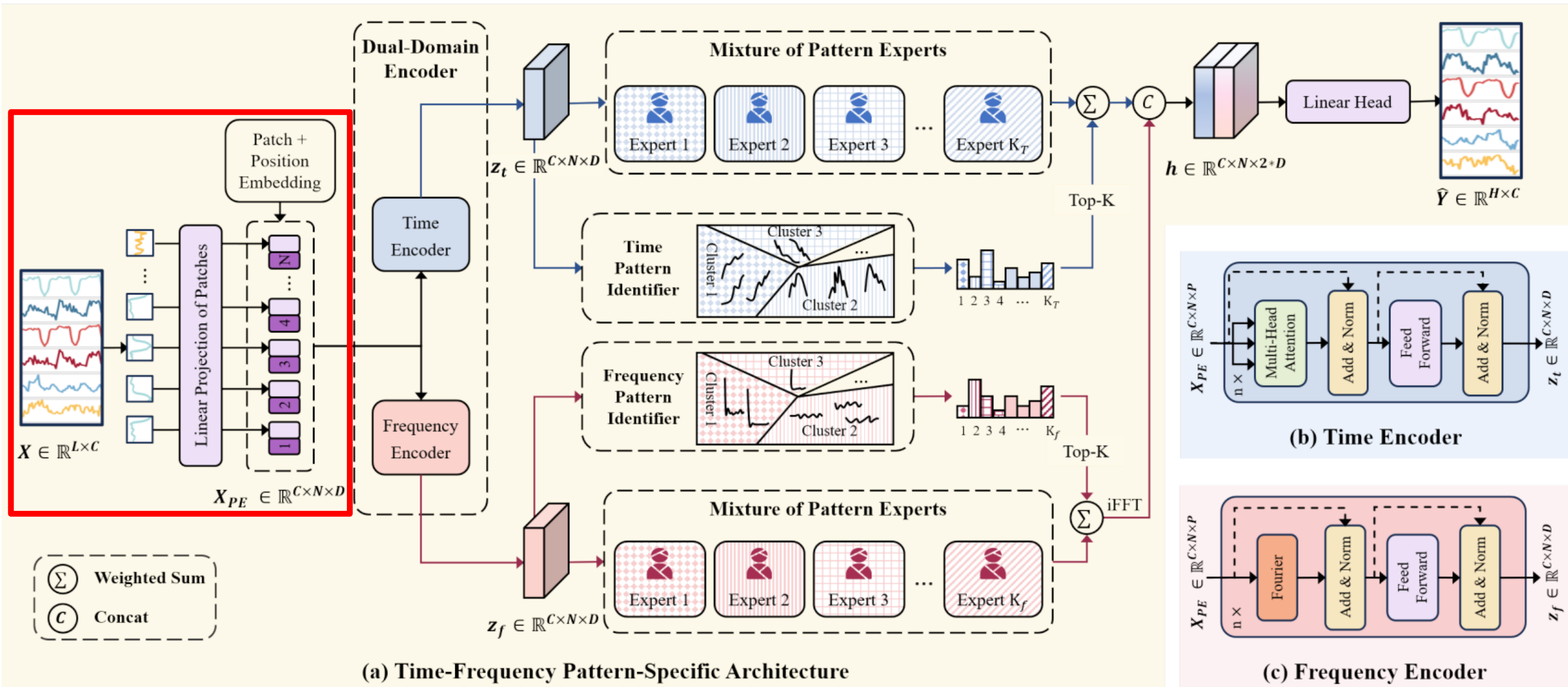


(a) Sudden drift

(b) Gradual drift

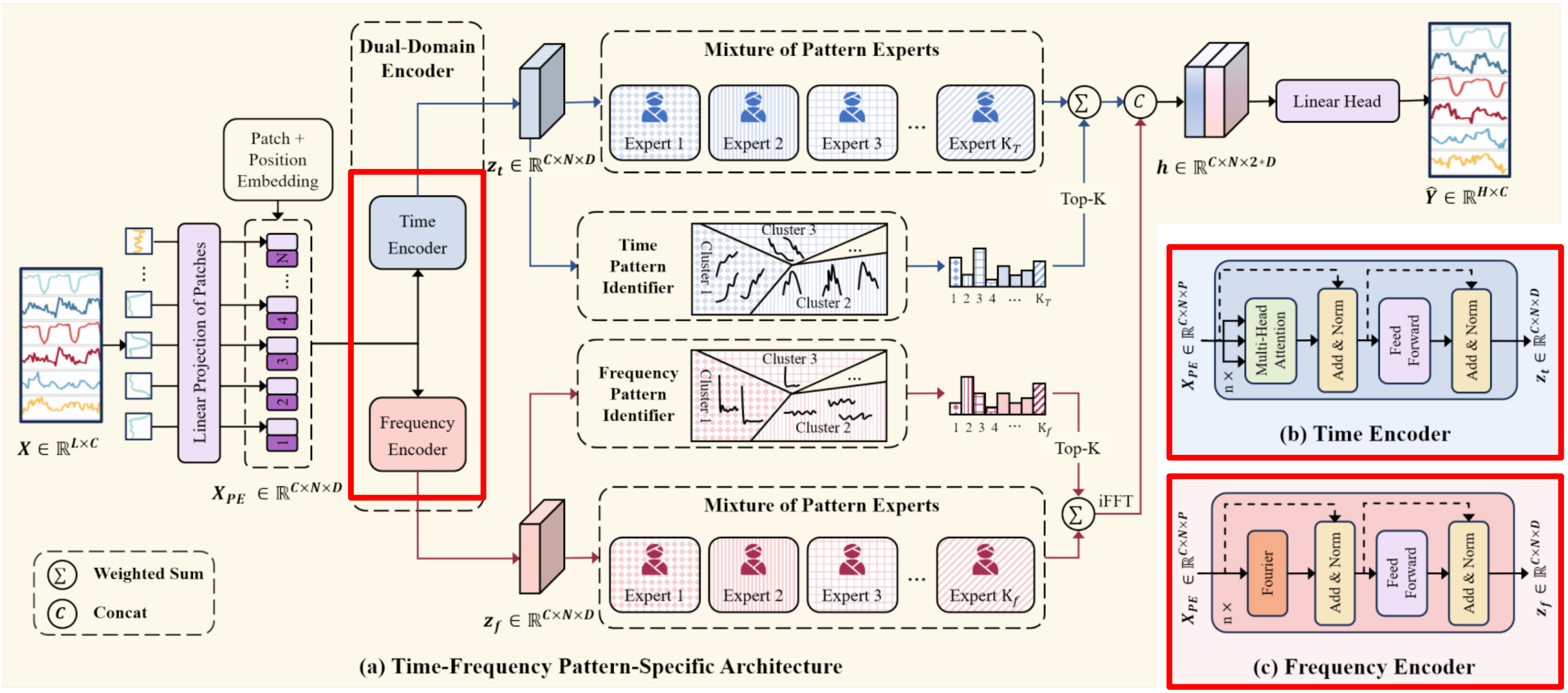
- 기존 시계열 예측 모델들은 모든 데이터를 단일 분포에서 나온 샘플로 취급하는 균등 분포 모델링 방식을 사용
  실제 데이터의 Distribution Shift 문제 발생 시 예측 성능이 하락함
- TFPS의 Time-Frequency Domain에서 각 Patch들의 특징 추출 및 부분 군집화를 통해 데이터의 패턴을 동적으로 식별
  이후 각 패턴에 최적화된 Experts를 통해 복잡한 시계열 변동을 효과적으로 포착

모델 구조



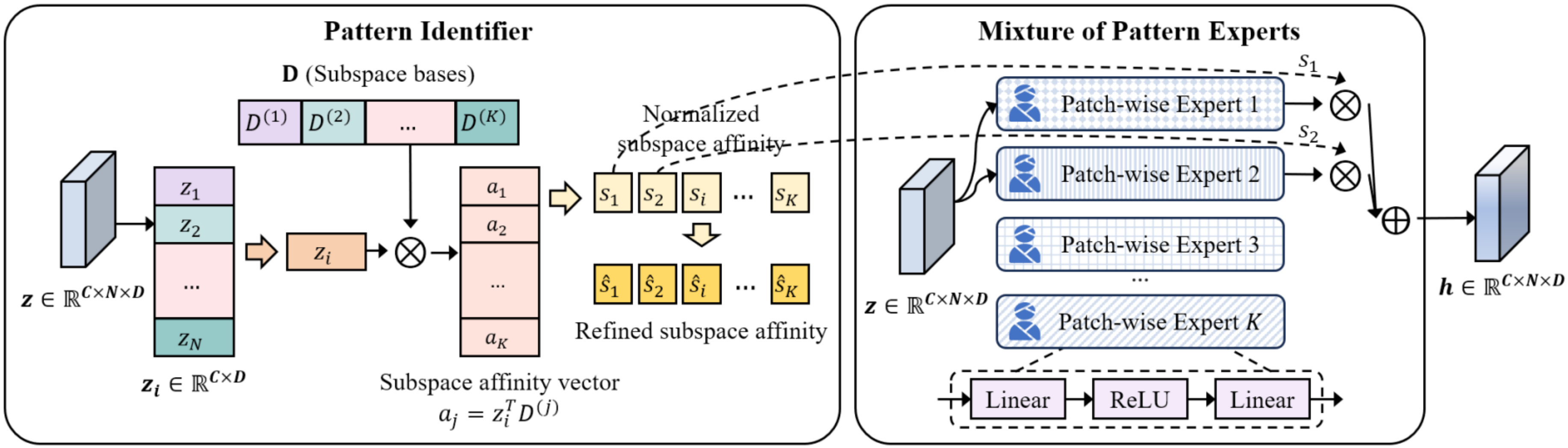
- Patch & Linear Projection
  - Channel Independece 방식으로, 각 채널을 일정 길이의 Patch로 나누고, Linear Projection으로 D차원 임베딩 벡터로 변환
  - 이후 임베딩 벡터는 복사되어 각각의 Encoder에 동일하게 입력

모델 구조



- Time Encoder
  - 시계열의 전반적인 추세와 장기, 단기 의존성 파악
  - Multi-Head Attention 연산을 통해 시간 특징 벡터 $z_t \in \mathbb{R}^{C \times N \times D}$ 생성
- Frequency Encoder
  - 시계열의 주기성을 파악
  - Self-Attention 대신 Fast Fourier Transform을 통해 주파수 특징 벡터 $z_f \in \mathbb{R}^{C \times N \times D}$ 생성

모델 구조



- Pattern Identifier
  - 입력 Patch z_i와 Learnable Parameter D 간의 행렬곱 수행
  - 행렬곱 결과에 정규화 적용하여 현재 Patch가 각 패턴과 얼마나 닮았는지 0~1 사이의 값으로 변환, 이후 Top-K개의 높은 점수만 남기고 나머지는 0으로 처리해 특정 패턴을 명확하게 함
- Mixture of Pattern Experts
  - 특징 벡터를 각 Expert에 입력하여 나온 출력값과 Top-K개의 유사도를 행렬곱, 이후 나온 값을 가중합하여 최종 Pattern-Specific Representation 생성

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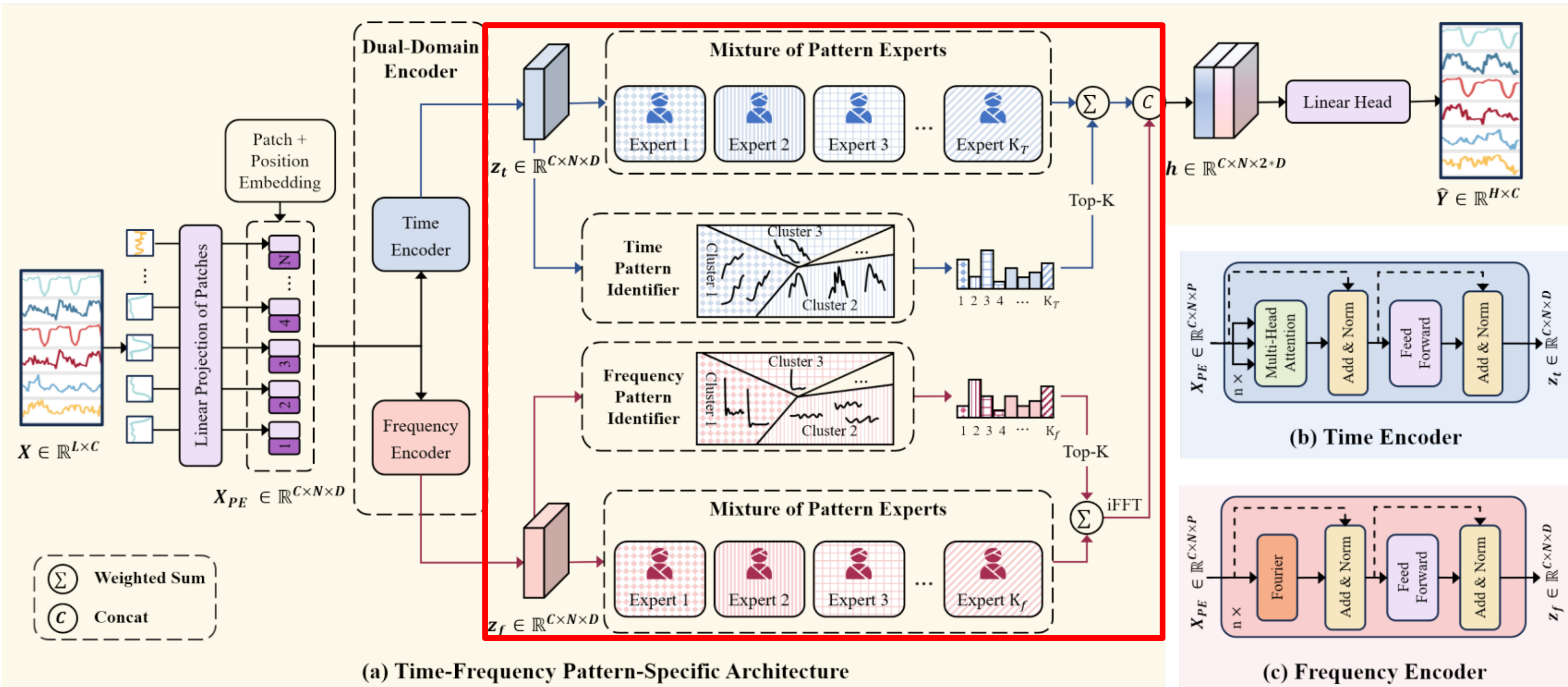

- Pattern Identifier
  - 입력 Patch z_i와 Learnable Parameter D의 유사도 측정 후, Top-K개의 높은 점수만 남김
- Mixture of Pattern Experts
  - 각 Expert의 출력 벡터와 Top-K개의 유사도를 행렬곱, 이후 출력값을 가중합하여 최종 Representation $h \in \mathbb{R}^{C \times N \times D}$ 생성
  - Time Encoder와 Frequency Encoder의 최종 Representation을 Concat하고 Linear Head를 통해 최종 예측값 생성

주요 모델 성능 비교

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Model		IMP.	TFPS (Our)		TSLANet (2024)		FITS (2024)		iTransformer (2024)		TFDNet-IK (2023)		PatchTST (2023)		TimesNet (2023)		DLinear (2023)		FEDformer (2022)	
Metric		MSE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	-1.1%	0.398	0.413	0.387	0.405	0.395	**0.403**	0.387	0.405	0.396	0.409	0.413	0.419	0.389	0.412	0.398	0.410	**0.385**	0.425
	192	4.8%	**0.423**	**0.423**	0.448	0.436	0.445	0.432	0.441	0.436	0.451	0.441	0.460	0.445	0.441	0.442	0.434	0.427	0.441	0.461
	336	1.8%	**0.484**	**0.461**	0.491	0.487	0.489	0.463	0.491	0.463	0.495	0.462	0.497	0.463	0.491	0.467	0.499	0.477	0.491	0.473
	720	3.0%	**0.488**	**0.476**	0.505	0.486	0.496	0.485	0.509	0.494	0.492	0.482	0.501	0.486	0.512	0.491	0.508	0.503	0.501	0.499
ETTh2	96	-2.0%	0.313	0.355	0.290	0.345	0.295	0.344	0.301	0.350	**0.289**	**0.337**	0.299	0.348	0.324	0.368	0.315	0.374	0.342	0.383
	192	-2.9%	0.405	0.410	**0.362**	**0.391**	0.382	0.396	0.380	0.399	0.379	0.395	0.383	0.398	0.393	0.410	0.432	0.447	0.434	0.440
	336	10.5%	**0.392**	**0.415**	0.401	0.419	0.416	0.425	0.424	0.432	0.416	0.422	0.424	0.431	0.429	0.437	0.486	0.481	0.512	0.497
	720	12.6%	**0.410**	**0.433**	0.419	0.439	0.418	0.437	0.430	0.447	0.424	0.441	0.429	0.445	0.433	0.448	0.732	0.614	0.467	0.476
ETTm1	96	4.1%	**0.327**	**0.367**	0.329	0.368	0.354	0.375	0.342	0.377	0.331	0.369	0.331	0.370	0.337	0.377	0.346	0.374	0.360	0.406
	192	2.6%	**0.374**	0.395	0.376	0.383	0.392	0.393	0.383	0.396	0.376	**0.381**	0.374	0.395	0.395	0.406	0.382	0.392	0.395	0.427
	336	4.2%	**0.401**	**0.408**	0.403	0.414	0.425	0.415	0.418	0.418	0.405	0.410	0.402	0.412	0.433	0.432	0.414	0.414	0.448	0.458
	720	-0.7%	0.479	0.456	**0.445**	0.438	0.486	0.449	0.487	0.457	0.471	**0.437**	0.466	0.446	0.484	0.458	0.478	0.455	0.491	0.479
ETTm2	96	6.9%	**0.170**	**0.255**	0.179	0.261	0.183	0.266	0.186	0.272	0.176	0.267	0.177	0.260	0.182	0.262	0.184	0.276	0.193	0.285
	192	7.1%	**0.235**	**0.296**	0.243	0.303	0.247	0.305	0.254	0.314	0.245	0.302	0.248	0.306	0.252	0.307	0.282	0.357	0.256	0.324
	336	4.6%	**0.297**	**0.335**	0.308	0.345	0.307	0.342	0.316	0.351	0.303	0.340	0.303	0.341	0.312	0.346	0.324	0.364	0.321	0.364
	720	3.6%	**0.401**	**0.397**	0.403	0.400	0.407	0.401	0.414	0.407	0.405	0.399	0.405	0.403	0.417	0.404	0.441	0.454	0.434	0.426
Exchange	96	12.7%	**0.083**	**0.205**	0.085	0.206	0.088	0.210	0.086	0.206	0.084	0.205	0.089	0.206	0.105	0.233	0.089	0.219	0.136	0.265
	192	11.2%	**0.174**	**0.297**	0.178	0.300	0.181	0.304	0.181	0.304	0.176	0.299	0.178	0.302	0.219	0.342	0.180	0.319	0.279	0.384
	336	10.4%	**0.310**	**0.398**	0.329	0.415	0.324	0.413	0.338	0.422	0.321	0.409	0.326	0.411	0.353	0.433	0.313	0.423	0.465	0.504
	720	-13.3%	1.011	0.756	0.850	0.693	0.846	0.696	0.853	0.696	**0.835**	**0.689**	0.840	0.690	0.912	0.724	0.837	0.690	1.169	0.826
Weather	96	15.6%	**0.154**	**0.202**	0.176	0.216	0.167	0.214	0.176	0.216	0.165	0.209	0.177	0.219	0.168	0.218	0.197	0.257	0.236	0.325
	192	10.6%	**0.205**	**0.249**	0.226	0.258	0.215	0.257	0.225	0.257	0.214	0.252	0.225	0.259	0.226	0.267	0.237	0.294	0.268	0.337
	336	9.1%	**0.262**	**0.289**	0.279	0.299	0.270	0.299	0.281	0.299	0.267	0.298	0.278	0.298	0.283	0.305	0.283	0.332	0.366	0.402
	720	4.1%	**0.344**	**0.342**	0.355	0.355	0.347	0.345	0.358	0.350	0.347	0.346	0.351	0.346	0.355	0.353	0.347	0.382	0.407	0.422
Electricity	96	14.6%	**0.149**	**0.236**	0.155	0.249	0.200	0.278	0.151	0.241	0.171	0.254	0.166	0.252	0.168	0.272	0.195	0.277	0.189	0.304
	192	12.0%	**0.162**	**0.253**	0.170	0.264	0.200	0.281	0.167	0.258	0.189	0.269	0.174	0.261	0.186	0.289	0.194	0.281	0.198	0.312
	336	0.2%	0.200	0.310	0.197	0.282	0.214	0.295	**0.179**	**0.271**	0.205	0.284	0.190	0.277	0.197	0.298	0.207	0.296	0.212	0.326
	720	7.2%	**0.220**	0.320	0.224	0.318	0.256	0.328	0.229	0.319	0.247	0.318	0.230	**0.312**	0.225	0.322	0.243	0.330	0.242	0.351
Traffic	96	21.1%	**0.427**	0.296	0.475	0.307	0.651	0.388	0.428	**0.295**	0.519	0.314	0.446	0.284	0.586	0.316	0.650	0.397	0.575	0.357
	192	17.7%	**0.445**	0.298	0.478	0.306	0.603	0.364	0.448	0.302	0.513	0.314	0.453	0.285	0.618	0.323	0.600	0.372	0.613	0.381
	336	17.0%	**0.459**	**0.307**	0.494	0.312	0.610	0.366	0.465	0.311	0.525	0.319	0.467	0.291	0.634	0.337	0.606	0.374	0.622	0.380
	720	15.1%	**0.496**	**0.313**	0.528	0.331	0.648	0.387	0.501	0.333	0.561	0.336	0.501	0.492	0.659	0.349	0.646	0.396	0.630	0.383
ILI	24	40.9%	**1.349**	**0.760**	1.749	0.898	3.489	1.373	2.443	1.078	1.824	0.824	1.614	0.835	1.699	0.871	2.239	1.041	3.217	1.246
	36	43.6%	**1.239**	**0.752**	1.754	0.912	3.530	1.370	2.455	1.086	1.699	0.813	1.475	0.859	1.733	0.913	2.238	1.049	2.688	1.074
	48	40.4%	**1.461**	**0.801**	2.050	0.984	3.671	1.391	3.437	1.331	1.762	0.831	1.642	0.880	2.272	0.999	2.252	1.064	2.540	1.057
	60	39.8%	**1.458**	**0.836**	2.240	1.039	4.030	1.462	2.734	1.155	1.758	0.863	1.608	0.885	1.998	0.974	2.236	1.057	2.782	1.136
1st Count			57		3		1		3		6		1		0		0		1	

Ablation Study

Table 2: Ablation study of TFPS components. The model variants in our ablation study include the following configurations across both time and frequency branches: (a) inclusion of the encoder, PI and MoPE; (b) PI replaced with Linear; (c) only the encoder. The best results are in **bold**.

Time Branch			Frequency Branch			ETTh1				ETTh2			
Encoder	PI	MoPE	Encoder	PI	MoPE	96	192	336	720	96	192	336	720
✓	✓	✓	✓	✓	✓	**0.398**	**0.423**	**0.484**	**0.488**	**0.313**	**0.405**	**0.392**	**0.410**
✓	✓	✓				0.401	0.459	0.486	0.492	0.318	0.409	0.400	0.428
✓	Linear	✓				0.401	0.451	0.494	0.509	0.325	0.411	0.400	0.434
✓						0.414	0.460	0.501	0.500	0.339	0.411	0.426	0.431
			✓	✓	✓	0.455	0.507	0.539	0.576	0.324	0.407	0.417	0.436
			✓	Linear	✓	0.503	0.535	0.558	0.583	0.398	0.446	0.457	0.444
			✓			0.552	0.583	0.591	0.594	0.371	0.426	0.418	0.463

- Time Branch와 Frequency Branch를 모두 사용한 기존 모델이 가장 좋은 성능을 보임
- Time Branch가 모델 성능에 큰 영향을 끼치는 것을 확인

이후 계획

- 논문 재현 실험 및 아이디어 구상 후 적용 실험

부록